



Public Transit User Dynamics: A Simulation Framework Under Crowding Situation in [location] Context

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Background study

General problem

Rationale of the study

Research objective

Research question

Research gap and contribution

Future task

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Rationale of the study

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General Problem



Source: TransJakarta Ltd., 2025

- Emerging cities have been increasingly promoting the use of public transportation.
- This trend has led to a growing demand for public transport systems, particularly metro rail and city bus services, within urban contexts.
- Instead of non-peak hour, passengers demand intensifies during peak hour serving various trip purposes such as commuting to work and school.
- Accordingly, critical processes such as passenger movement/dynamics within alighting and boarding during the public transport operations can serve as key contributors to the operations..
- This kind of specific process has been extensively modeled by various approaches such as social force model approach (Yining Jia, et.al.,2026, Shen et.al., 2026, Li et.al., 2020), Markov process (Baali et.al., 2025).

Background study

General problem

Rationale of the study

Research objective

Research question

Research gap and contribution

Future task

Rationale

- Although boarding-alighting assessment studies have gained attention, since this critical process can affect public transport performance, most prior studies have primarily focused on passenger movements associated with entering and exiting vehicles during boarding and alighting processes (Qu Y., et.al.,2019, Seriani and Fernandez, 2015).
- At this moment, only Yining Jia, et.al.(2026) explicitly consider a categorization of metro passengers into three groups: **boarding**, **alighting**, and **remaining on-board passengers**.
- However, their analysis is situated within normal operating conditions, where remaining passengers on-board are typically seated, using experimental settings by Fu L., et.al., (2023).
- The focus of this study is alighting and boarding process in crowding situation, the categorization of the passenger type is needed and remain underexplored when examine this process, as individual interaction between passenger might occur during alighting and boarding process.
- Aligned with previous study, the behavior of each pedestrian is different, since it severely influences the movement characteristics, thus, developing crowd model by implementing multiple pedestrian classes is essential (Duives D.C.,Daamen W., Hoogendoorn S.P., 2013).

Rationale



Source: TransJakarta Ltd., 2025

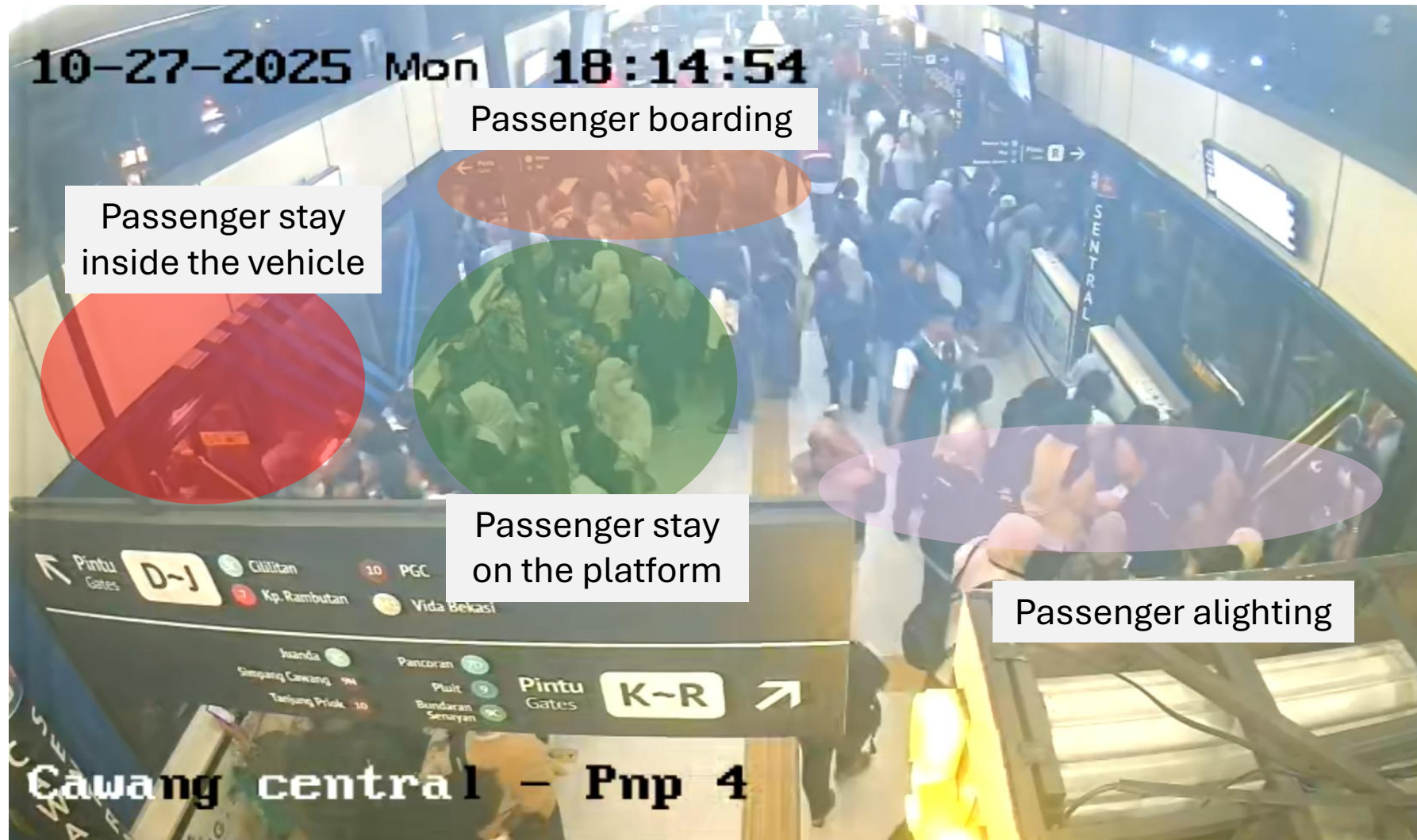
- On the other hand, related to the approach of side, Seriani and Fernandez, (2015) argued that continuous movement on pedestrian dynamics modelling ensure more realistic representations as the space is free from artificial restrictions, such as cells or grids in Cellular Automata approach.
- However, continuous model seem to be able to reproduce pedestrian movement, but the movements of the pedestrians are not directly related to decision-making processes of pedestrians (Asano, Iryo & Kuwahara, 2009)

Rationale

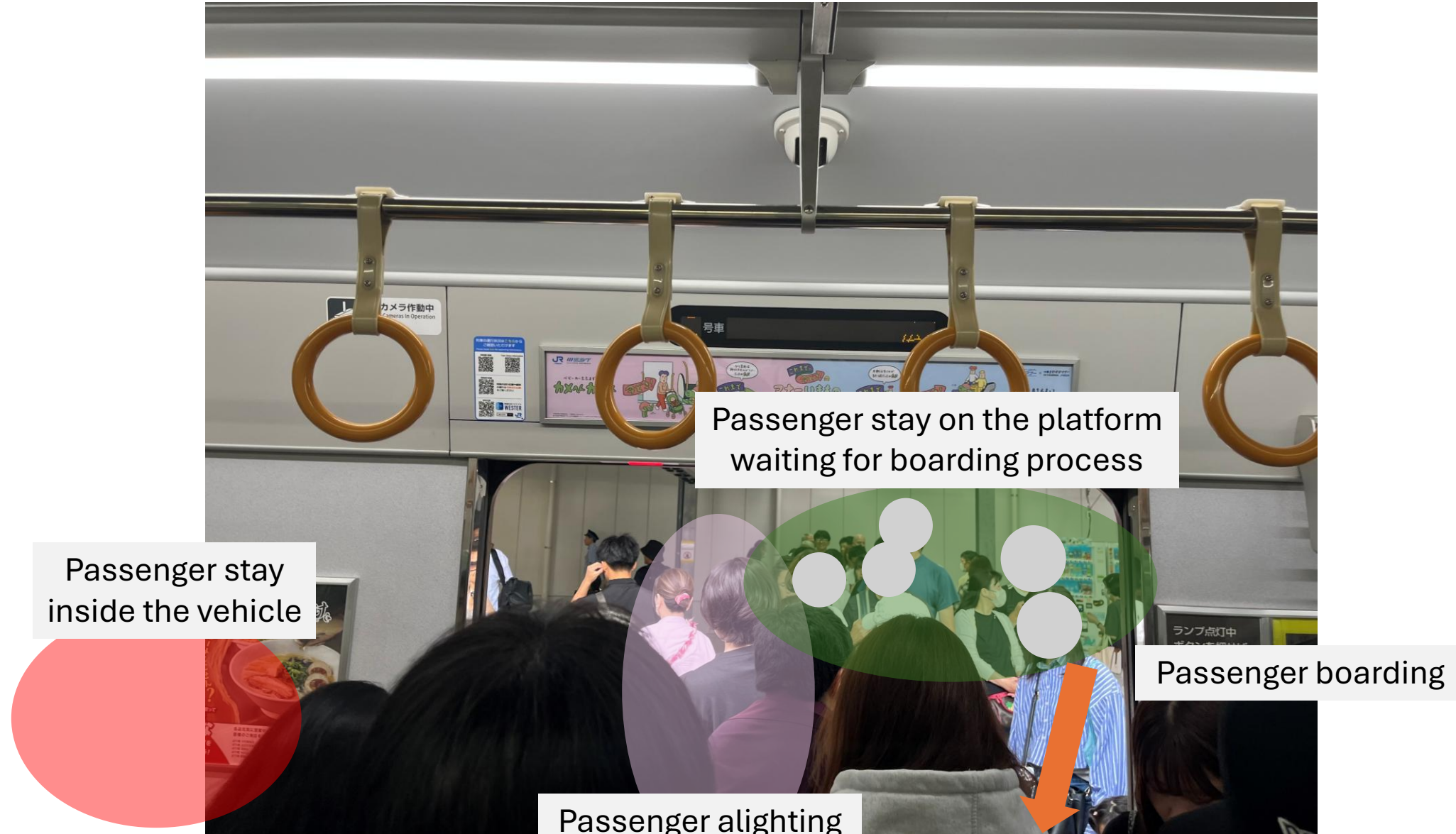
- The study proposes the passenger category, during the boarding and alighting process, depends on each state, are described as follows:

State	Type	Represented by
Non-stationary	Passenger alighting	PA_m
	Passenger boarding	PB_m
Stationary	Passenger stay inside the car	PC_s
	Passenger stay on the platform waiting for boarding process	PP_s

Rationale



Rationale



Source: Personal documentation, 2026

Rationale

- The interaction might be occurred between each passenger type during boarding and alighting process are described as follows:

	Passenger stay inside the car/vehicle (PC_s)	Passenger stay on the platform waiting for boarding process (PP_s)	Passenger alighting (PA_m)	Passenger boarding (PB_m)
Passenger stay inside the car/vehicle (PC_s)	na	na	(5)	(9)
Passenger stay on the platform waiting for boarding process (PP_s)	na	na	(6)	similar
Passenger alighting (PA_m)	(1)	(3)	(2) And (4)	*
Passenger boarding (PB_m)	(7)	similar	*	(8)

* This interaction is unlikely to occur when the “Disembarking Precedes Embarking” rule is applied but may arise in cases of non-compliance.

Rationale

The interaction might be occurred between each passenger type during boarding and alighting process are described as follows:

Alighting

Passenger alighting (PA_m)

- (1) Collision avoidance with passengers inside the car since preparing to step out of car
- (2) Leader follower with another passenger alighting
- (3) Collision avoidance with passenger waiting at the platform preparing for boarding
- (4) Collision avoidance with another passenger alighting at the arriving platform

Passenger stay inside the car (PC_s)

- (5) Collision avoidance with passenger alighting when preparing for stepping out of the car

Passenger stay on the platform waiting for boarding process (PP_s)

- (6) Collision avoidance with passenger alighting stepping in to arriving platform.

Rationale

The interaction might be occurred between each passenger type during boarding and alighting process are described as follows:

Boarding process

Passenger boarding (PB_m)

(7) Collision avoidance with passenger inside the car when stepping into the car looking for space for stand.

(8) Leader-follower interaction within passenger boarding.

Passenger stay inside the car (PC_s)

(9) Collision avoidance since they must adjust the position when passenger boarding coming in.

Background study

General problem

Rationale of the study

Research objective

Research question

Research gap and contribution

Future task

Research Objective

The aim of this study is to examine passengers' behavior during public transport alighting and boarding with the consideration of non-stationary and stationary passengers

Background study

General problem

Rationale of the study

Research objective

Research question

Research gap and contribution

Future task

Research Question

(RQ1) How do passengers' attitudes and perceptions influence their movement behavior in crowded-situated public transport, in the perspective of stationary state and non-stationary state passenger?

(RQ2) What alternative framework can better represent both alighting and boarding process, with the consideration of different passenger type, non-stationary and stationary, incorporated to the examination during crowded-situated in public transport?

Background study

General problem

Rationale of the study

Research objective

Research question

Research gap and contribution

Future task

Research Gap and Contribution

Type of gap	Identified gap	Study approach	Academic contribution	Practical contribution
Empirical gap	Lack of empirical or simulation-based studies on passenger dynamics in public transport that incorporate passenger states (non-stationary and stationary).	Examines passenger dynamics under high-density conditions by incorporating different passenger states.	Provides new evidence on passenger behavior and system performance in crowded scenarios with the consideration of different passenger states.	By accounting for different passenger states, operators can design more efficient alighting and boarding procedures as well as improved crowd management strategies.
Theoretical gap	Existing pedestrian modeling frameworks remain limited in describing pedestrian dynamics in continuous space within a discrete choice framework with taking passenger heterogeneity into account, particularly in high density transit context	Develops a continuous-space pedestrian modeling framework that integrates behavioral decision-making by considering for passenger heterogeneity.	Establishes a state-aware modeling framework that could represent human interaction during public transport operation	

Background study

General problem

Rationale of the study

Research objective

Research question

Research gap and contribution

Future task

Future Task

- Expand methodological understanding of how to examine passenger dynamics while maintaining analysis in continuous space within a discrete-based framework, and accounting for the different passenger types as proposed
- After the fundamental principal confirmed with Sensei, discuss and decide how to conduct data collection within the proposed framework.

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Appreciate your feedback and discussion :D